ISR (intelligence, surveillance, reconnaissance) sensors within concealed bays (Figure 6.7).

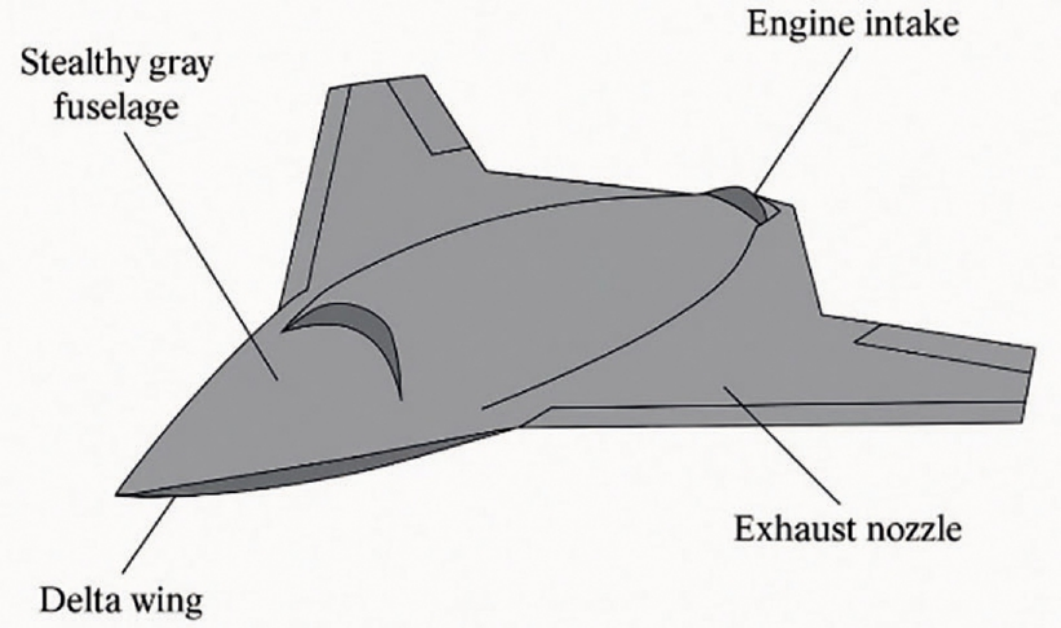


Figure 6.7 Boeing Phantom Ray (2011).⏎

Technologically, the Phantom Ray integrates AI-driven autonomous flight control systems, sensor fusion, and adaptive mission management software that allow it to operate with minimal human input. The UCAV can perform takeoff, navigation, mission execution, and landing autonomously, guided by an onboard system that processes environmental data in real time. This autonomy allows the Phantom Ray to execute dynamic mission adaptation, including route changes, threat avoidance, and coordinated operations with manned aircraft or other drones.

The propulsion system consists of a General Electric F404-GE-102D turbofan engine, enabling a maximum speed of Mach 0.85 and a service ceiling exceeding 40,000 feet. The aircraft's sleek, low-profile air intake and shielded exhaust contribute to its stealth profile by minimizing radar and heat emissions. In its 2011 test flights, the Phantom Ray demonstrated fully